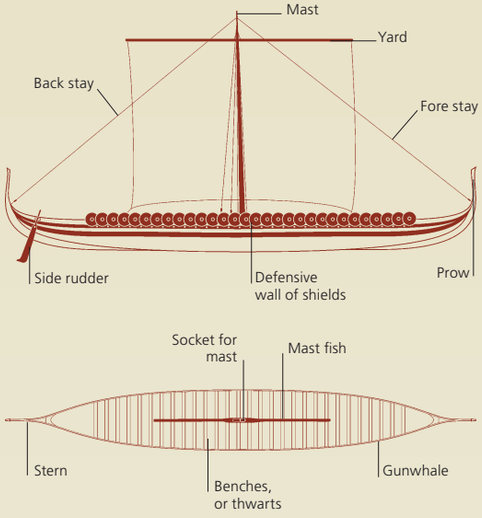
VIKING LONGSHIP

The Viking longship was a swift, sturdy, and versatile military transport. Propelled either by a sail or by oars, it could cross the open seas but also, because of its shallow draught, penetrate inland along rivers or be pulled up onto a beach. The longship featured here, known as *Havhingsten fra Glendalough* (the *Sea Stallion from Glendalough*), is a reconstruction of a vessel excavated from Roskilde Fjord, Denmark, in the 1960s.

The original of the *Sea Stallion*, referred to as *Skuldelev 2*, was deliberately sunk, along with four other ships, in the late 11th century. This was done to block the entrance of the fjord, thus protecting Roskilde, at that time the Danish capital, against attack from the sea. Study of the oak from which the vessel was made revealed that it had been built around 1042 in Dublin, Ireland, then a major Viking settlement.

The longship has a prow at both ends and is steered by means of a side rudder. It is clinker-built – that is, with a hull made of overlapping planks. As far as possible, the reconstruction used tools, techniques, and materials of the Viking period; the timber from 300 oak trees was required to build it. A longship of this size would have been commissioned by a man of wealth and power and its construction would have taken an entire winter.

At least 60 men would have been needed to row the longship. Their muscle power could have delivered a sustained speed of 5–6 knots, but with its sail raised and a favourable wind the ship could have made 15–20 knots.



Longship profile
The long, narrow shape of the longship was designed for speed. *Skuldelev 2* was approximately 30m (98ft) long but only 3.8m (12ft 6in) wide. It could sail in water less than a metre (3ft 3in) deep.



> NEVER BEFORE HAS SUCH TERROR APPEARED ... NOR WAS IT THOUGHT THAT SUCH AN INROAD FROM THE SEA COULD BE MADE.
>
> **ALCUIN** ON THE SACK OF LINDISFARNE, 793

Landfall
Vikings waited for a favourable wind before setting sail on a raid. They would get hardly any sleep on the gruelling voyage from England from Denmark, but could make it in about two days.



Shield
Vikings used their shields as a defence as they rowed to land on a hostile shore, but it is not known how they were attached to the side of the ship.



Clinker construction
The overlapping planks, or "strakes", are fixed with iron nails. The white objects are the locks sealing the oar ports. Only materials available to the Vikings were used in the linseed-oil paint.



Sail power and oar power
The rope fastened to the cleat *(left)* is one of the sheets that controls the yard (the wooden spar that carries the sail). When the ship was under sail and the oars not in use, the oar ports *(right)* were sealed with a special lock so that they would not let in water.



Oar port and frame timber
Oar ports were designed so that the whole oar and blade could pass through them. Frame timbers *(centre)* were fixed at intervals to the top three planks to reinforce the hull.